

Industrial Security (Bachelor)

Exercise Sheet – Section 02 ICS and SCADA Fundamentals

Exercise 2.1 – Component Mapping

Map each function below to the most appropriate component: **PLC, RTU, DCS, SCADA, HMI, Historian, SIS**.

1. Stores ten years of temperature readings at one-second resolution.
2. Brings a furnace to a safe state if pressure exceeds a threshold.
3. Runs ladder logic on the factory floor with a 10 ms cycle.
4. Displays a plant overview with active alarms to a night-shift operator.
5. Polls a remote pump station over a leased serial line.
6. Coordinates 5 000 control loops in a refinery from a unified vendor stack.

Exercise 2.2 – Ladder Logic

Design a ladder rung satisfying *all* of:

- A conveyor motor runs only when both safety guards (**Guard1, Guard2**) are closed.
- Pressing **Start** energises the motor; the motor latches.
- Pressing **Stop** or opening either guard immediately stops the motor.
- A pilot lamp lights whenever the motor is running.

Submit your solution as a ladder diagram with each contact labelled normally open (NO) or normally closed (NC).

Exercise 2.3 – Why an Independent SIS?

1. Explain in one paragraph why a Safety Instrumented System should be independent of the Basic Process Control System.
2. Sketch the TRITON attack from 2017: which system was targeted, what was the attacker's goal, why was the attack discovered?
3. List two design choices a vendor could make to harden a modern SIS against a TRITON-style attack.

Exercise 2.4 – Lab: OpenPLC

Complete `labs/lab-01-openplc`.

1. Install OpenPLC runtime + editor.
2. Implement the conveyor program from Exercise 2.2.
3. Connect a soft HMI (ScadaBR or Node-RED).
4. Trigger inputs and observe outputs.

5. Capture Modbus TCP traffic with Wireshark; identify the function codes.

Hint

Section 04 covers Modbus in detail; this lab will be referenced again then.